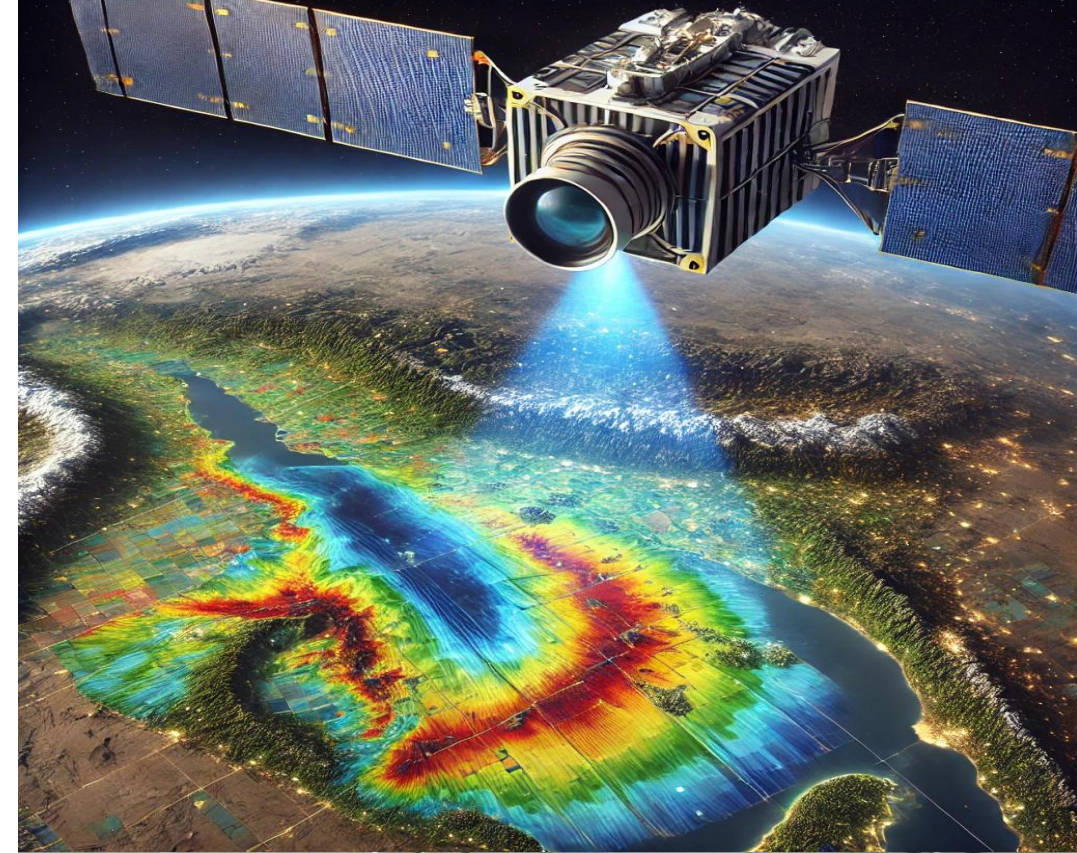
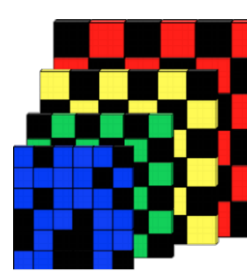
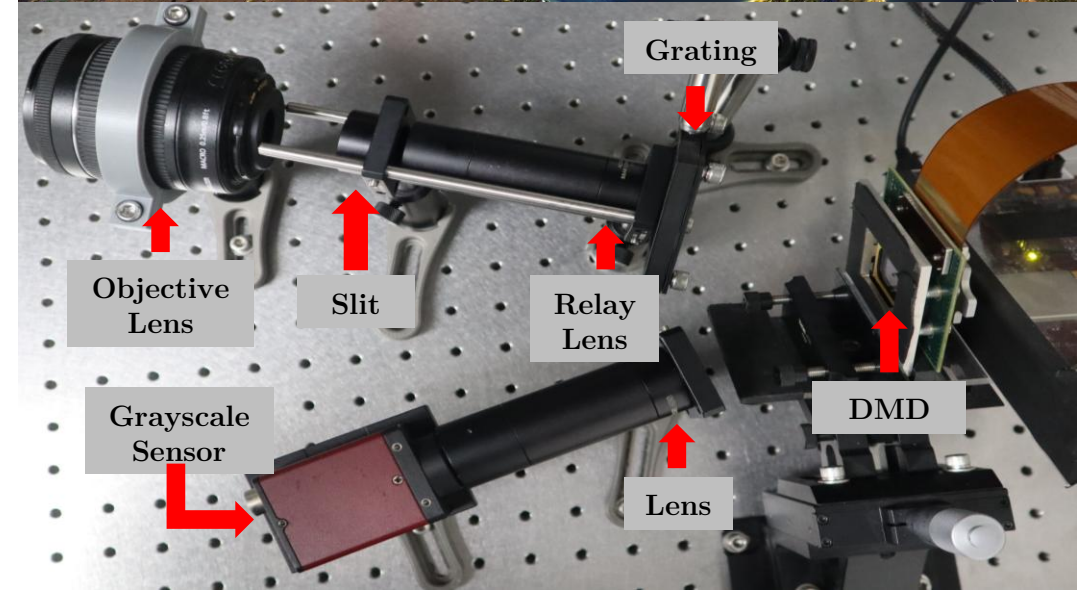




Bsc Electronic Engenieering
Msc in Electronic Engenieering
PhD (c) in Engenieering

Optical design, implementation and calibration of
hiperspectral imaging.
Design of Deep learning model for several task

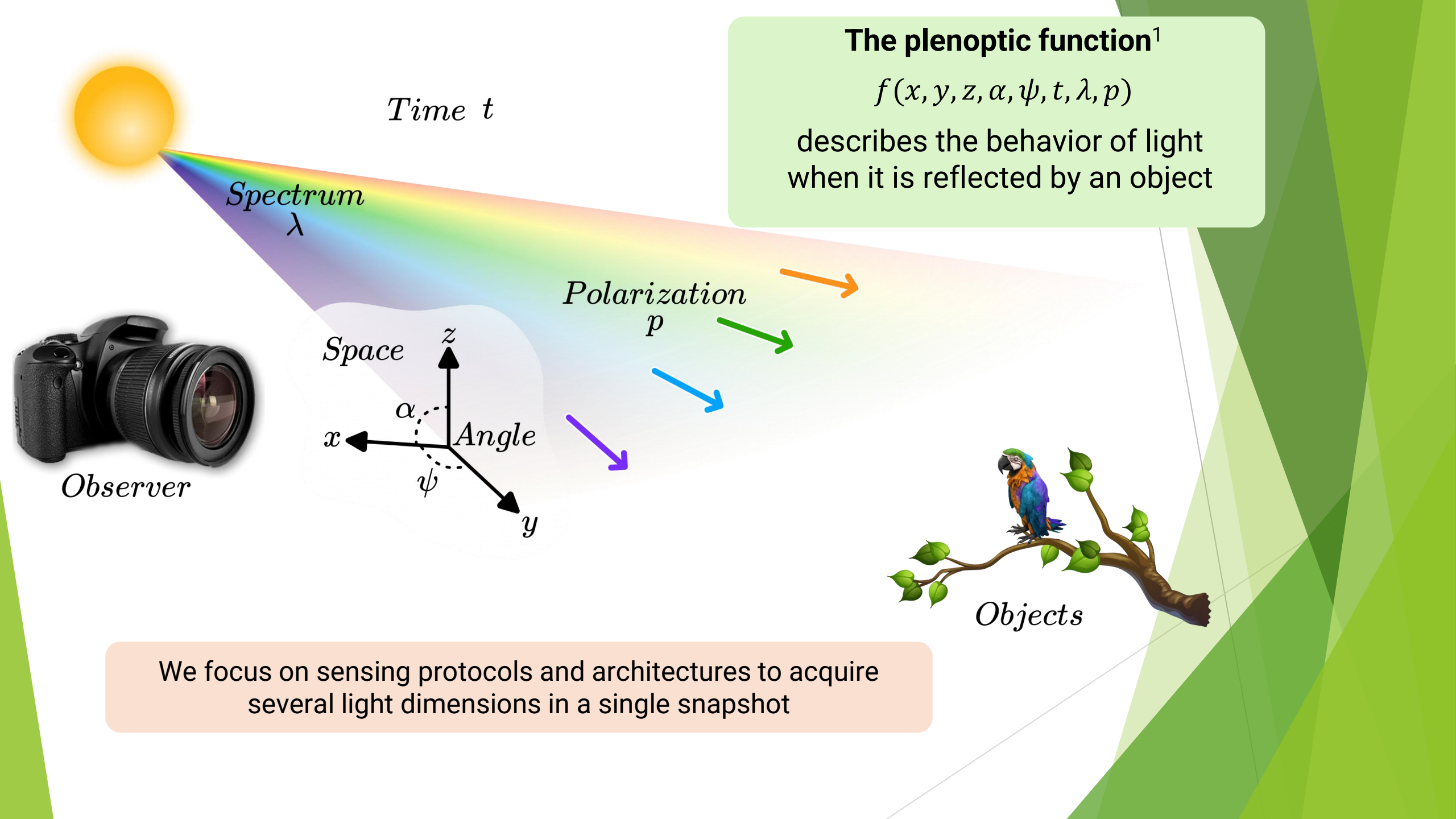


Contextualization

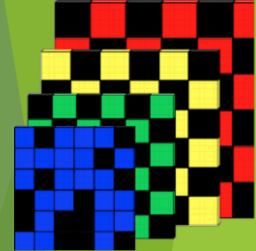
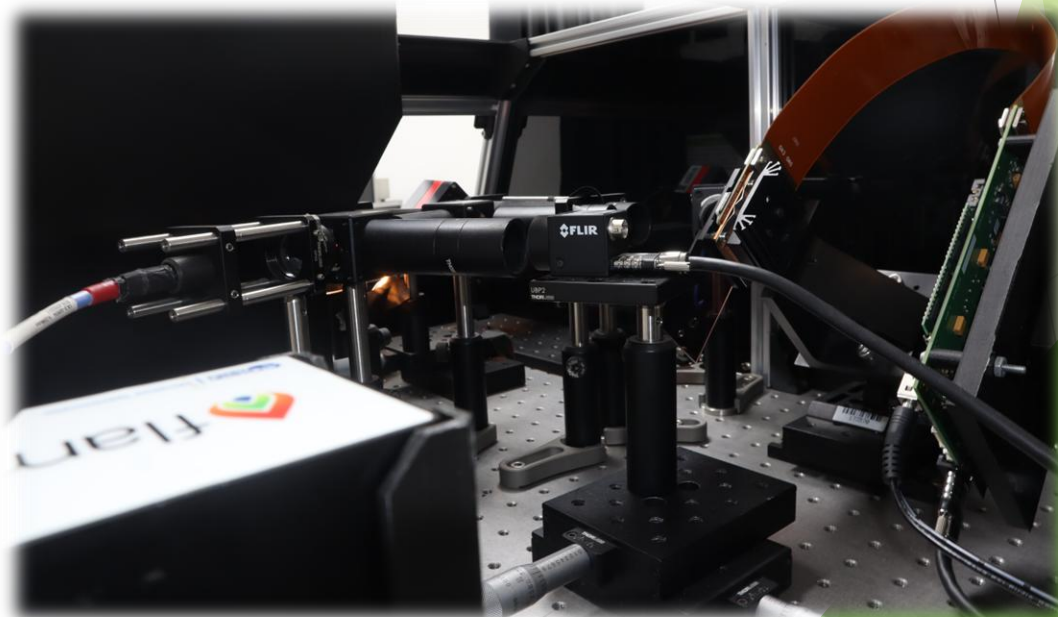
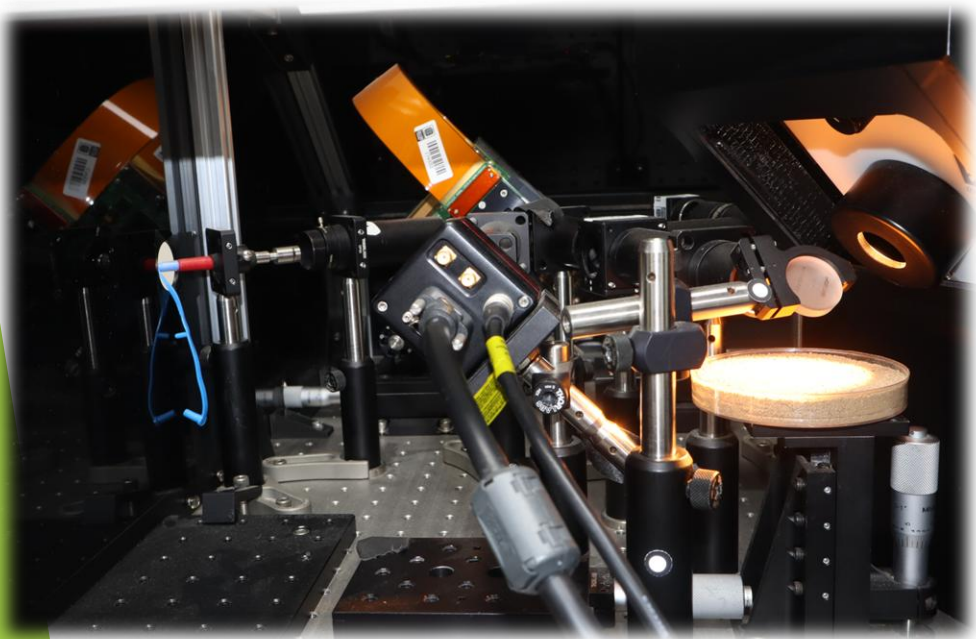
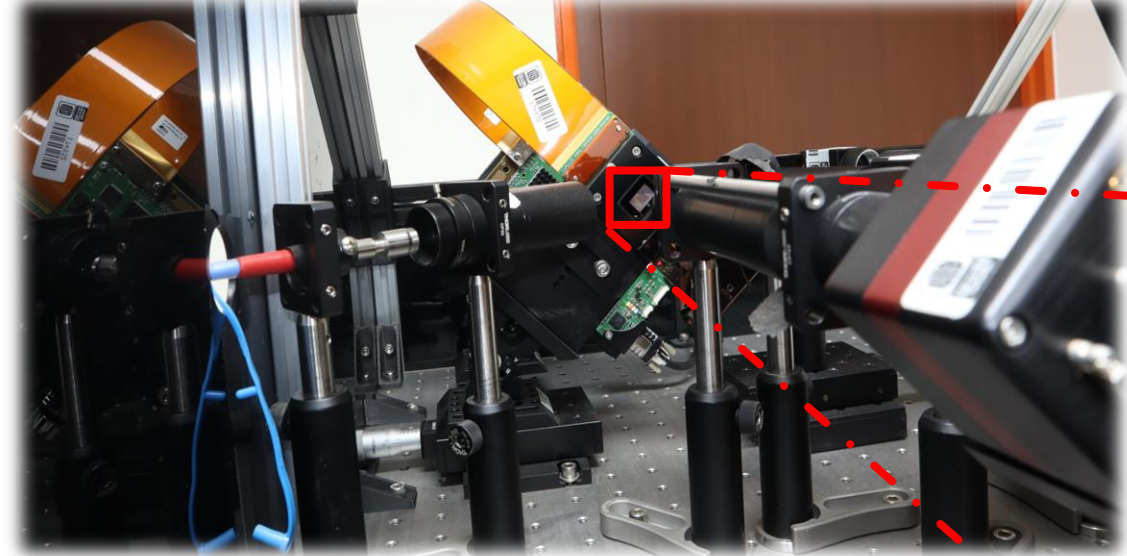
The plenoptic function¹

$$f(x, y, z, \alpha, \psi, t, \lambda, p)$$

describes the behavior of light when it is reflected by an object

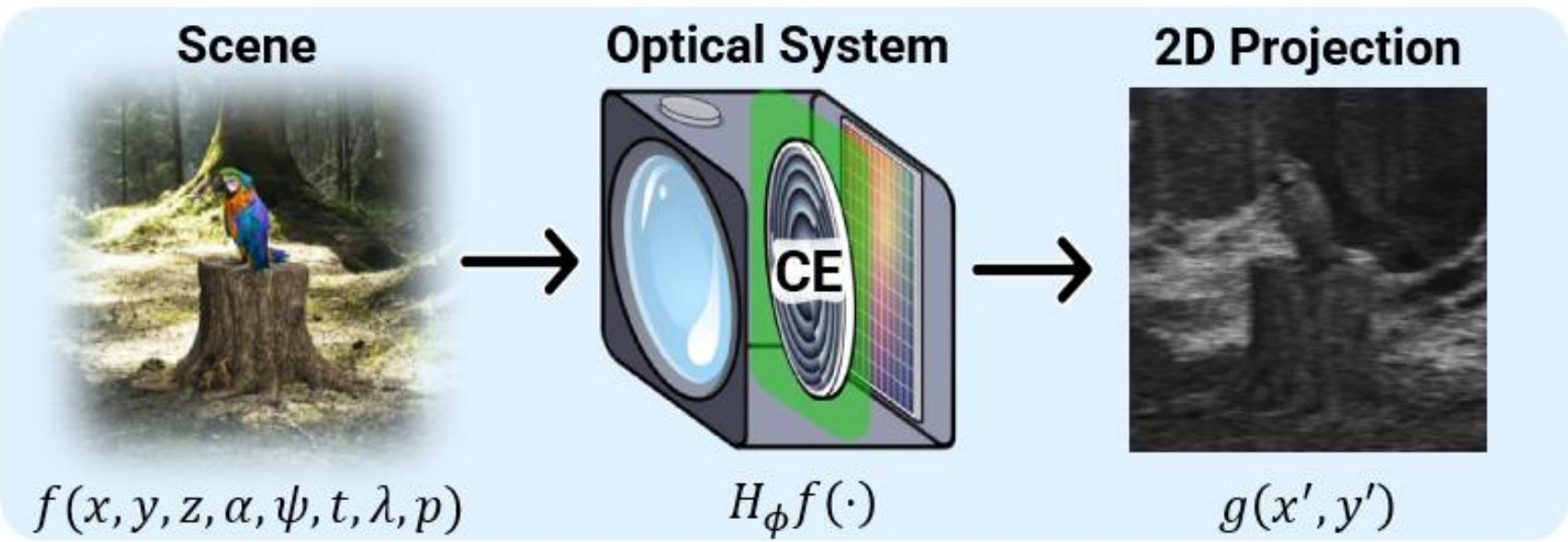


We focus on sensing protocols and architectures to acquire several light dimensions in a single snapshot



Computational Optical Imaging (COI)

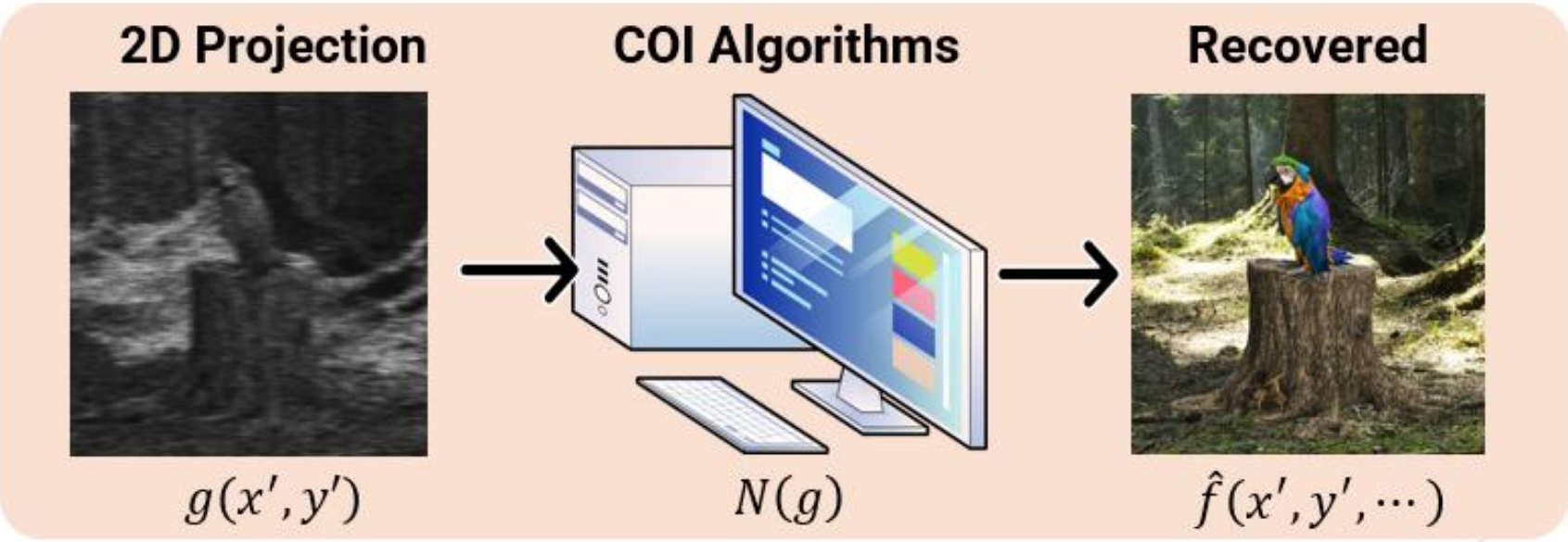
Acquisition



CE
Coded Elements

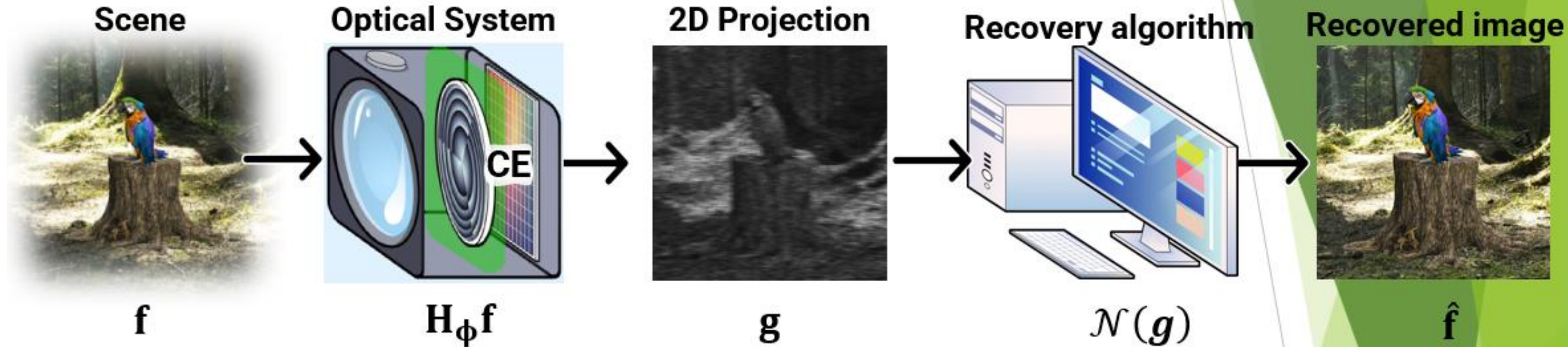
Low-cost acquisition can be done through specialized cameras and recovered through computational optical algorithms

Recovery



COI
Computational Optical Imaging

Computational Optical Imaging (COI)



Vectorized image

Additive noise

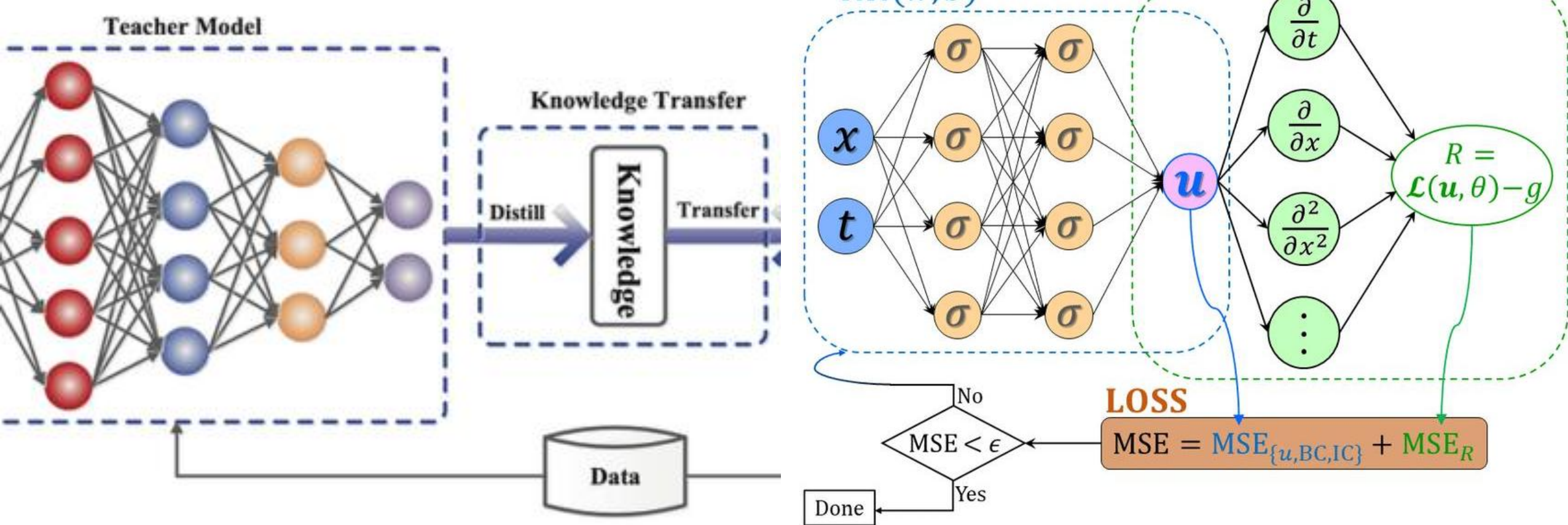
$$\mathbf{g} = \mathbf{H}_\phi \mathbf{f} + \omega$$

Vectorized measurements

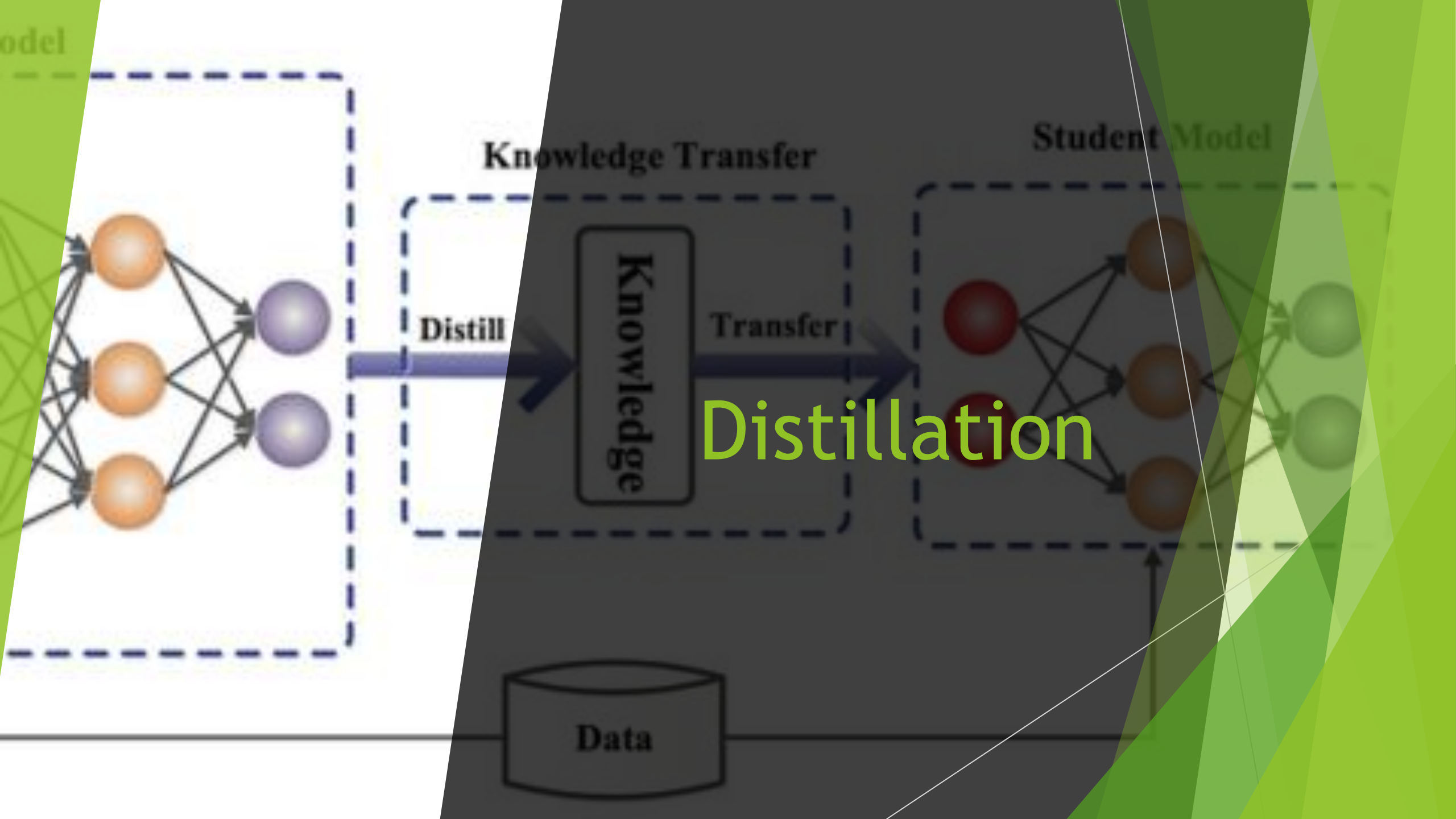
Sensing Matrix modeling the optics (encoder) containing the coding elements

We want to find the optimal encoder and decoder that minimizes the reconstruction error

$$\{\mathbf{H}_{\phi^*}, \mathcal{N}^*\} = \underset{\mathbf{H}_{\phi}, \mathcal{N}}{\operatorname{argmin}} \|\mathbf{f} - \mathcal{N}(\mathbf{H}_{\phi} \mathbf{f} + \omega)\|_2$$



Distillation and PINN concepts

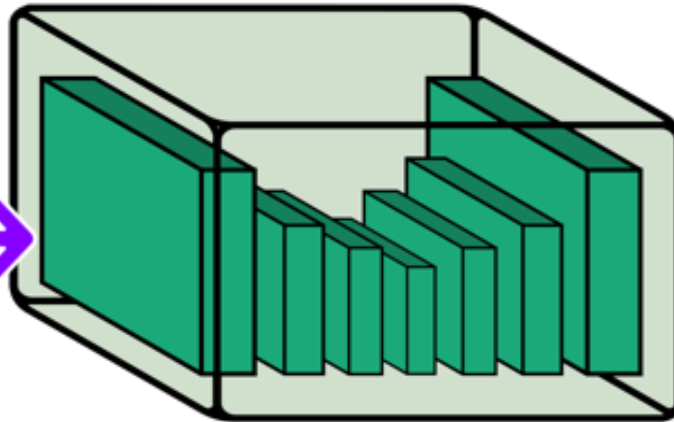


Problem Statement

Training data



Large neural network

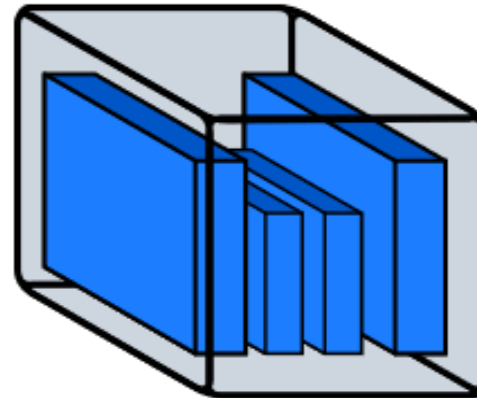


High performance on the task

Huge computational complexity

Massive storage requirements

Small neural network



Low performance on the task

Low computational complexity

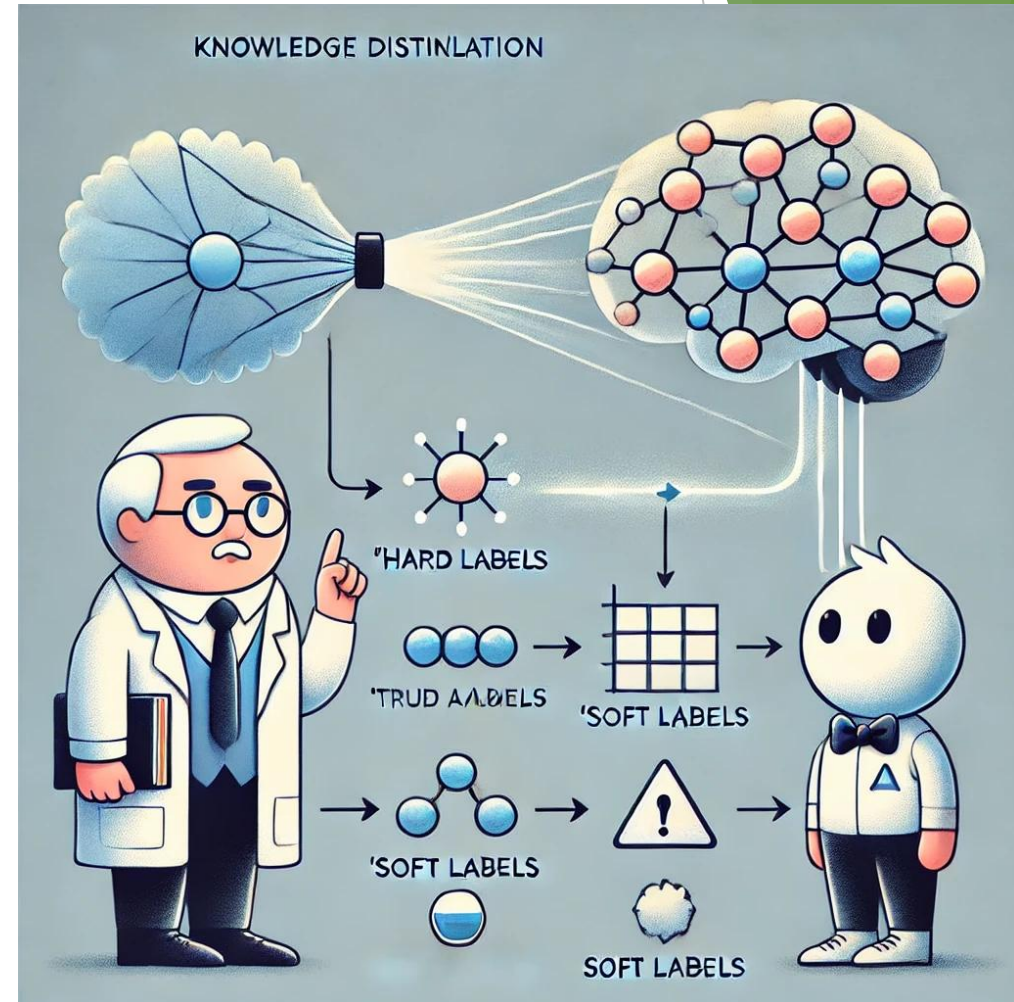
Minimal storage requirements

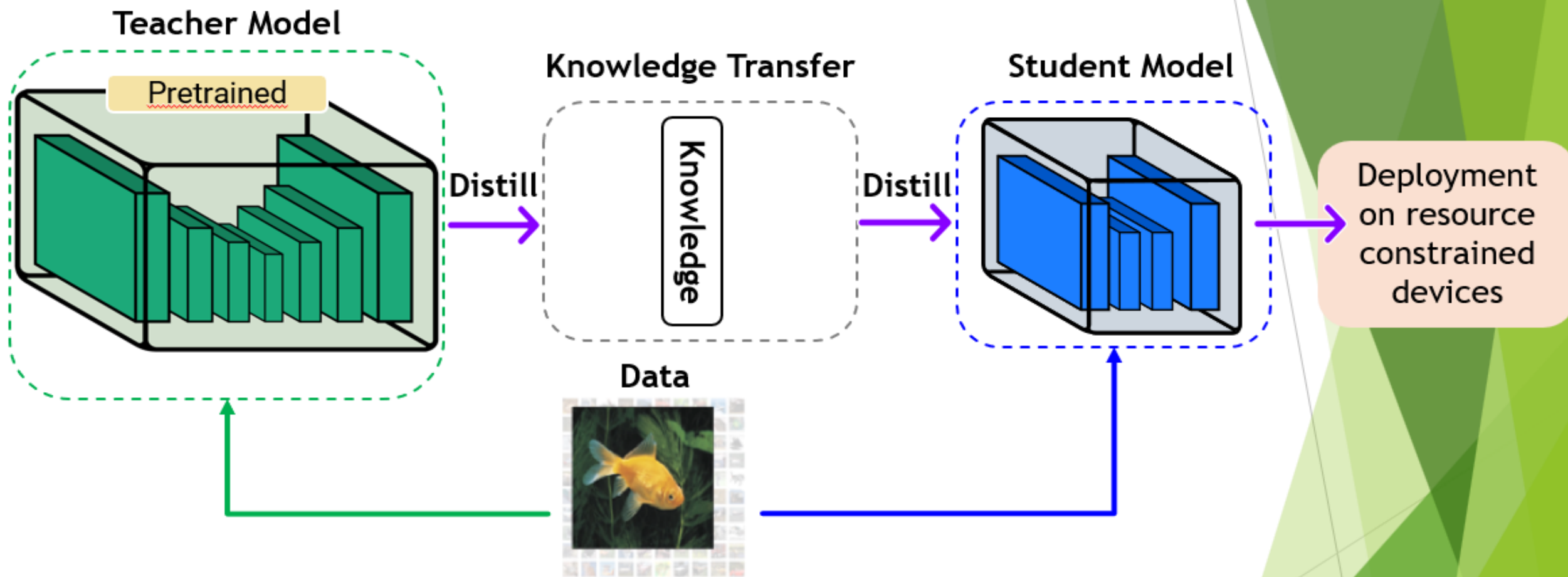
Deploying large neural networks on resource-constrained devices for real-time inference tasks is challenging due to their computational demands and memory requirements.

Distillation Concept



The concept originated from how humans actually learn: someone who knows a lot passes their knowledge to someone who knows little.





Knowledge distillation compresses a large neural network into a compact model, enabling deployment on resource-constrained devices

Knowledge Distillation Process

Train the Teacher Model



A large, high-accuracy model is trained on a dataset to learn complex patterns

Extracting Knowledge



The teacher generates two types of outputs for each input: hard labels and soft probabilities

Train the Student Model



A smaller, efficient model learns from the teacher's outputs while maintaining performance

Mathematical model

Given a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, we have:

- A **Teacher model** $f_T(x; \theta_T)$ with parameters θ_T .
- A **Student model** $f_S(x; \theta_S)$ with parameters θ_S .

Both models produce a logit output:

$$z_T = f_T(x; \theta_T), \quad z_S = f_S(x; \theta_S)$$

Where:

- $z_T \in \mathbb{R}^C$ and $z_S \in \mathbb{R}^C$ are logits for C classes.
- A **softmax with temperature** is applied to get probability distributions:

$$p_T = \sigma_T(z_T) = \frac{\exp(z_T/T)}{\sum_j \exp(z_{T,j}/T)}$$

$$p_S = \sigma_T(z_S) = \frac{\exp(z_S/T)}{\sum_j \exp(z_{S,j}/T)}$$

Where T is the **temperature** hyperparameter.

Student Loss Function

The Student minimizes a combination of two loss functions:

$$\mathcal{L}_{\text{Student}} = (1 - \lambda)\mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{Distil}}$$

1. Cross-Entropy (CE) Loss with ground truth labels y :

$$\mathcal{L}_{\text{CE}} = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[- \sum_j y_j \log \sigma(z_S) \right]$$

2. Distillation Loss (KL Divergence) between Teacher and Student:

$$\mathcal{L}_{\text{Distil}} = T^2 \cdot \mathbb{E}_{(x,y) \sim \mathcal{D}} [D_{\text{KL}}(p_T || p_S)]$$

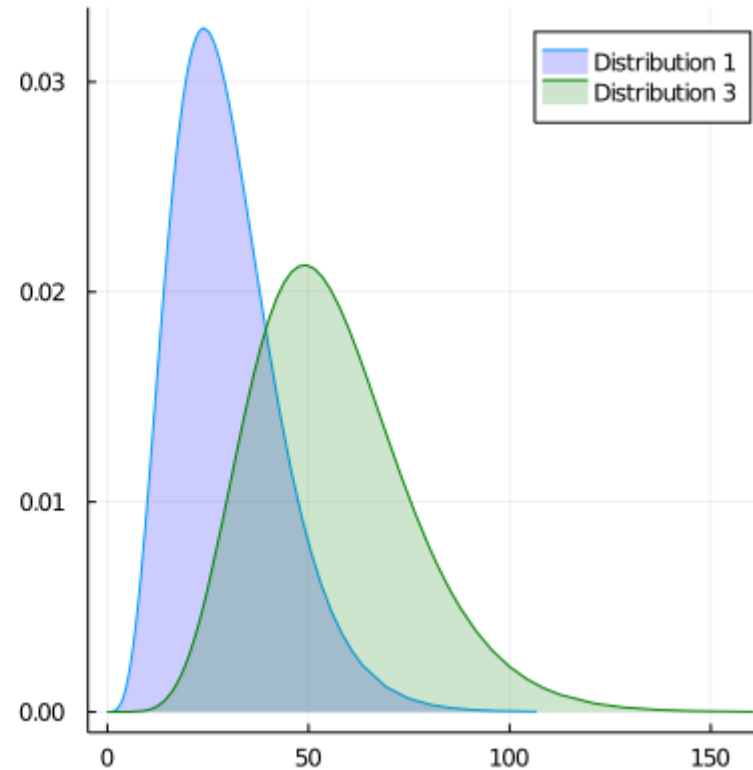
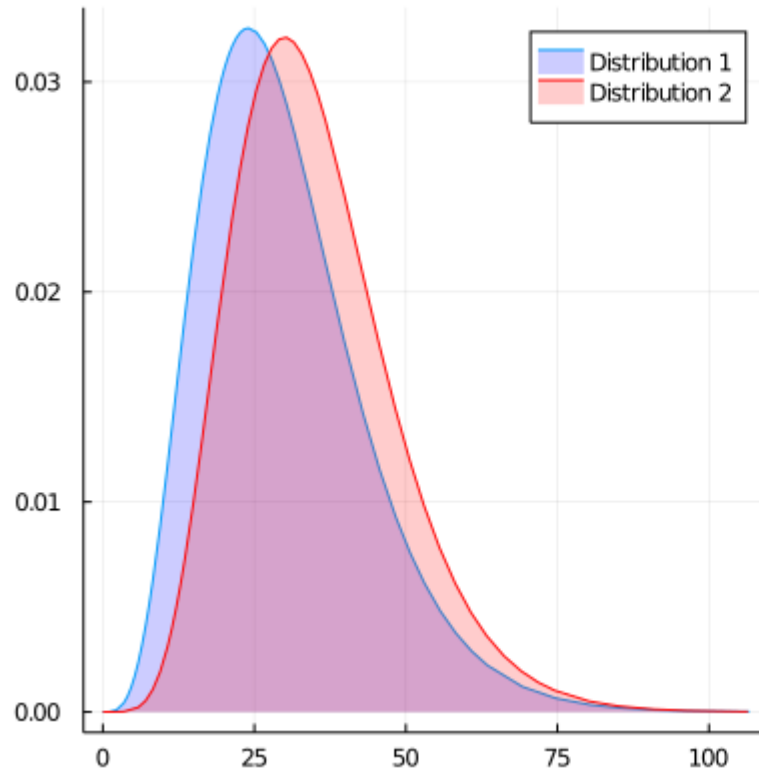
The T^2 factor ensures gradient stability during training.

$$D_{\text{KL}}(p_T || p_S) = \sum_j p_{T,j} \log \frac{p_{T,j}}{p_{S,j}}$$

Student Loss Function: KL Divergence

Is a **measure** of how **one probability distribution diverges** from a **second expected probability distribution**. It is a way of comparing two probability distributions, often a “true” distribution and an approximation of that distribution.

$$D_{\text{KL}}(p_T || p_S) = \sum_j p_{T,j} \log \frac{p_{T,j}}{p_{S,j}}$$



Student Loss Function: Interpretation

$$\mathcal{L}_{\text{Student}} = (1 - \lambda)\mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{Distil}}$$

- If $\lambda = 1$: The Student learns only from the Teacher's knowledge.
- If $\lambda = 0$: The Student learns only from ground truth labels.
- If $0 < \lambda < 1$: The Student combines both, balancing real data and the Teacher's knowledge.

Teacher Loss Function

The Teacher model is trained using a traditional **Cross-Entropy Loss** on the labeled dataset:

$$\mathcal{L}_{\text{Teacher}} = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[- \sum_j y_j \log p_T(j|x) \right]$$

where:

- x is the input sample.
- y is the true one-hot encoded label.
- $p_T(j|x)$ is the probability distribution of class j , obtained via **softmax** on the Teacher's logits z_T :

Teacher Loss: Special Cases

While the standard approach is **Cross-Entropy Loss**, some variations exist:

1. When the Teacher is Pre-trained (Fixed Model)

- No additional loss; the Teacher is frozen and used only for distillation.

2. When the Teacher is Co-trained with the Student (Self-Distillation)

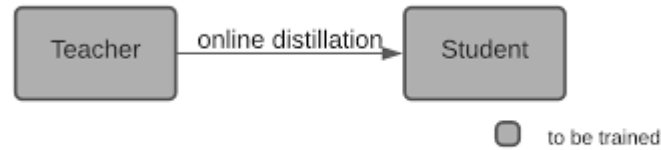
- The Teacher can be trained with an additional **distillation loss** to improve soft labels.

Types of Distillation

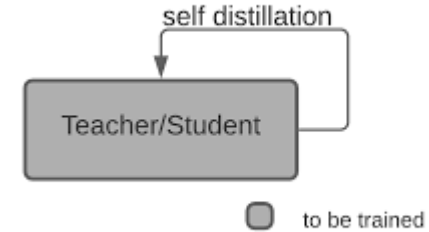
Offline distillation



Online distillation



Self-distillation

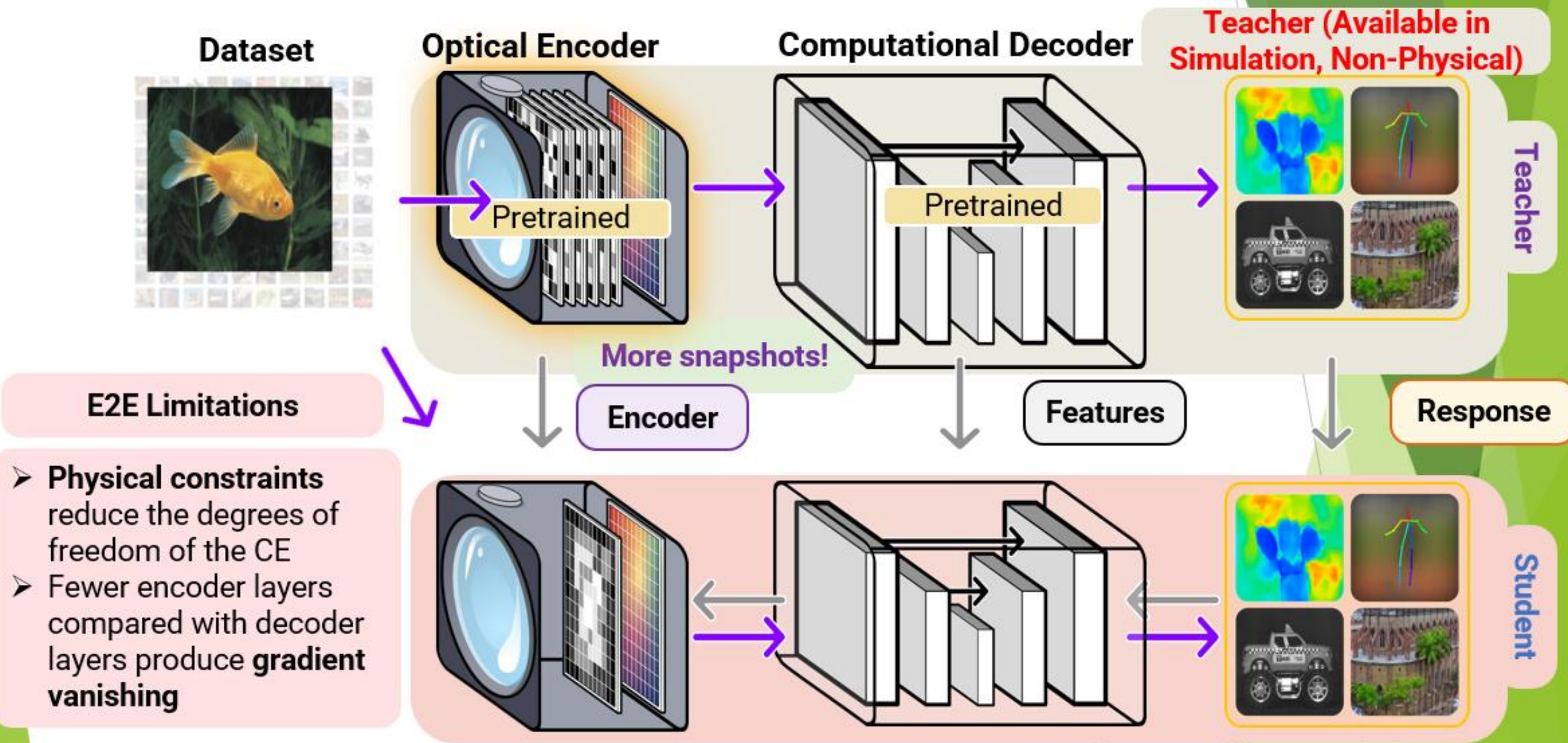


The most popular model distillation method, which involves pre-training a teacher model, **freezing it**, and later using it to **train a student model**. During the model distillation process, the training focuses on the student model as the teacher model remains unmodified.

Online distillation is an **end-to-end technique** in which the student and teacher models are trained simultaneously. In other words, the **teacher model is continuously** updated with new datasets, and the **student model is trained to reflect these changes**.

Involves using the **same networks for the teacher and student** models. In this case, a model learns from itself, meaning knowledge from the deeper layers of the model is continuously distilled into its shallow section. This method **solves** the conventional **limitations of discrepancy** in accuracy **between teacher and student models**.

Better than E2E



Applications

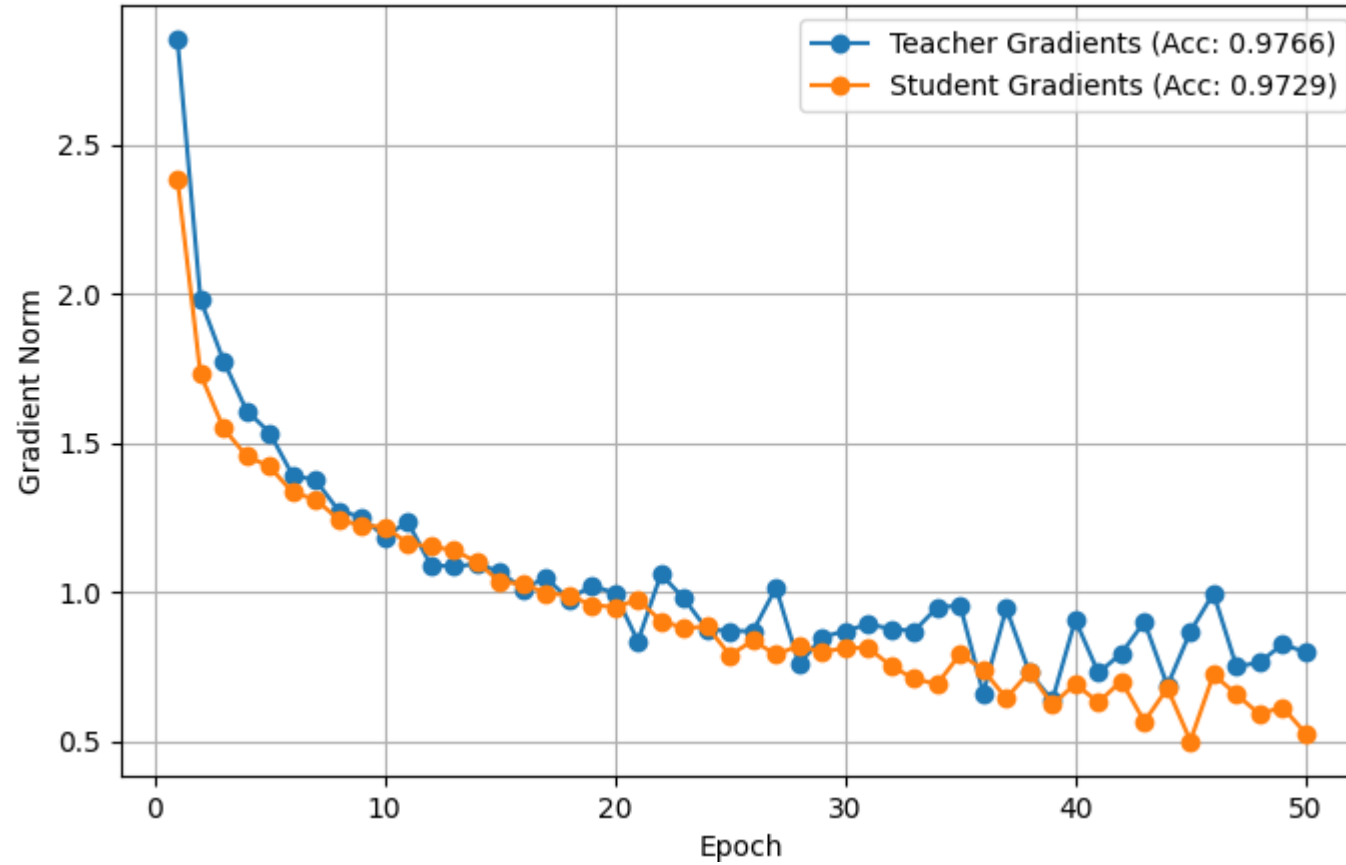


Convert heavy models with excellent performance to compressed models with comparable performance but which can be used with fewer computational resources.



Code

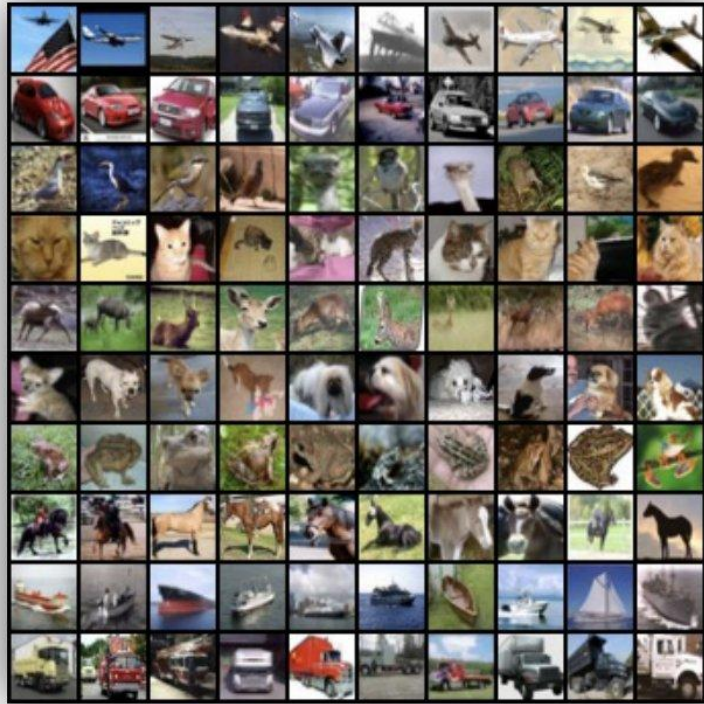
Comparación de Gradientes y Performance: Teacher vs Student



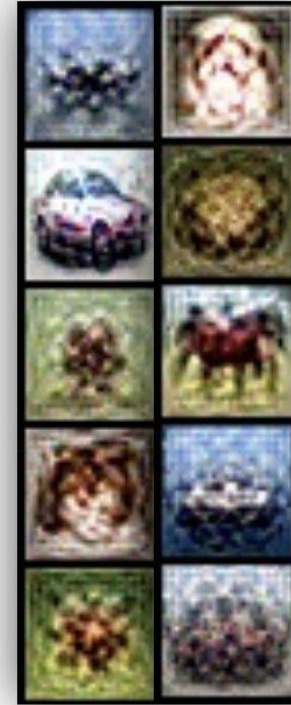
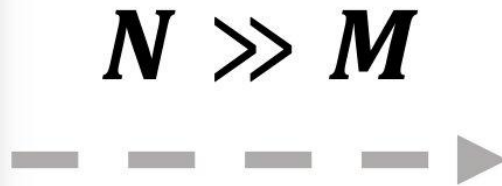
Performance	Accuracy
Teacher	0,991
Student	0,969

Special Case: Dataset Distillation

Problem Statement



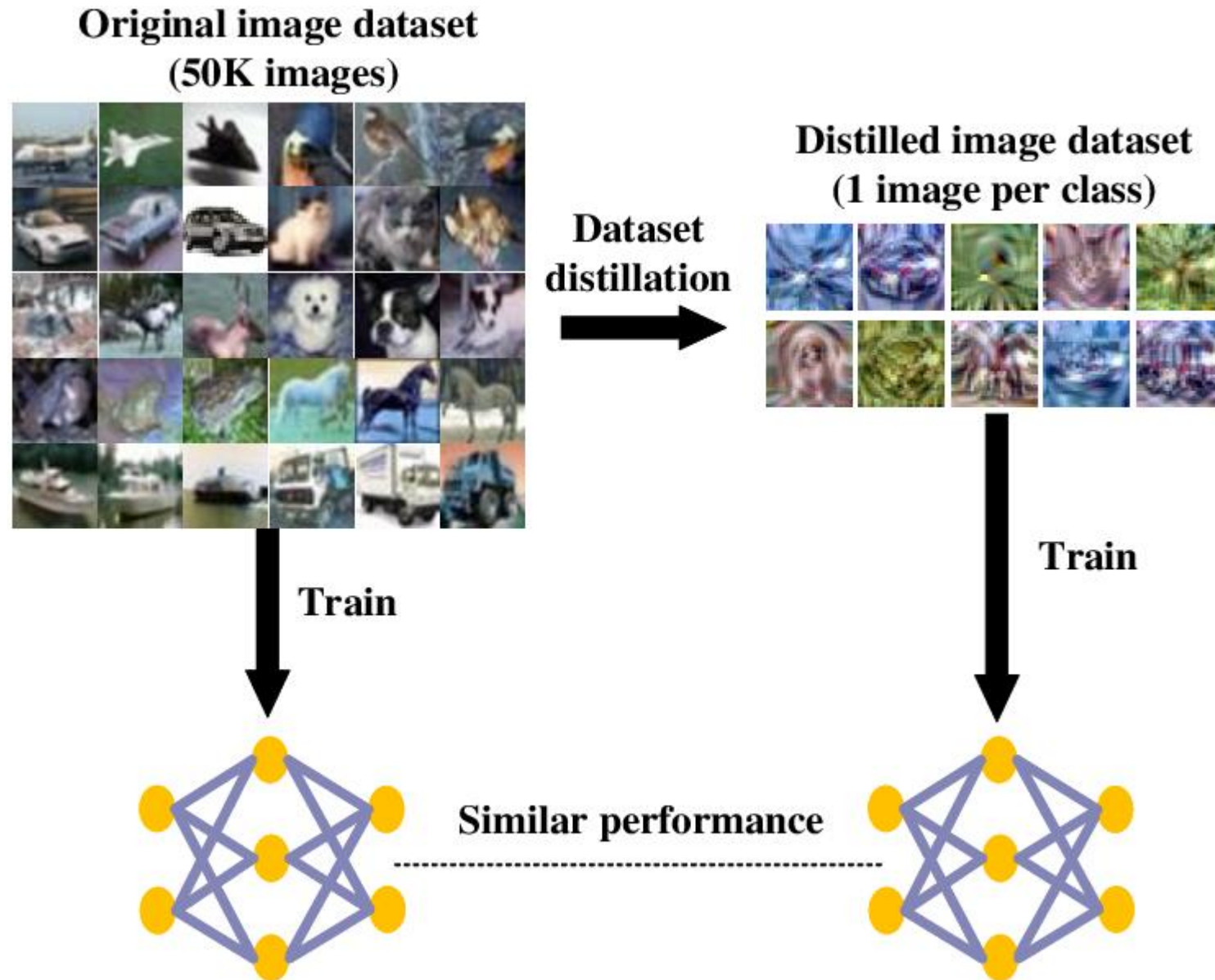
N images



M images

Sometimes the problem is not that the model is too large, but that the data needed to obtain a good result is too much.

Main Idea



Mathematical Model

Dataset distillation learns a synthetic dataset D_{syn} by optimizing:

$$D_{\text{syn}} = \arg \min_{D'} \mathbb{E}_{\theta_S \sim \mathcal{A}(D')} [\mathcal{L}(\theta_S, D)]$$

where:

- $\mathcal{A}(D')$ is the **training procedure** on the synthetic dataset D' .
- $\mathcal{L}(\theta_S, D)$ is the **loss function** measuring how well a student model trained on D' generalizes to the full dataset D .

Mathematical Model

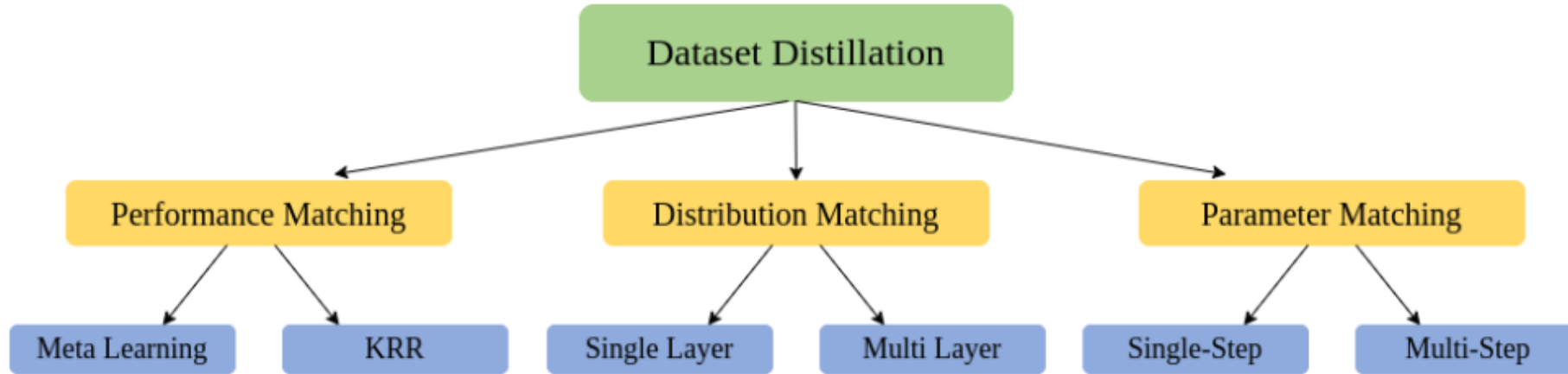
- The synthetic dataset is optimized such that training on it produces **similar gradients** to training on the full dataset:

$$\|\nabla_{\theta}\mathcal{L}(\theta, D) - \nabla_{\theta}\mathcal{L}(\theta, D_{\text{syn}})\|$$

The goal of **dataset distillation** is to generate a synthetic dataset **D_{syn}** such that training a model on it produces gradients similar to those obtained with the original dataset **D** .

This is achieved by minimizing the difference between the gradients of the loss function **$\mathcal{L}$** computed on **D** and **D_{syn}** , allowing models trained on **D_{syn}** to achieve performance similar to training on **D** , but with significantly fewer data.

Types of Dataset Distillation



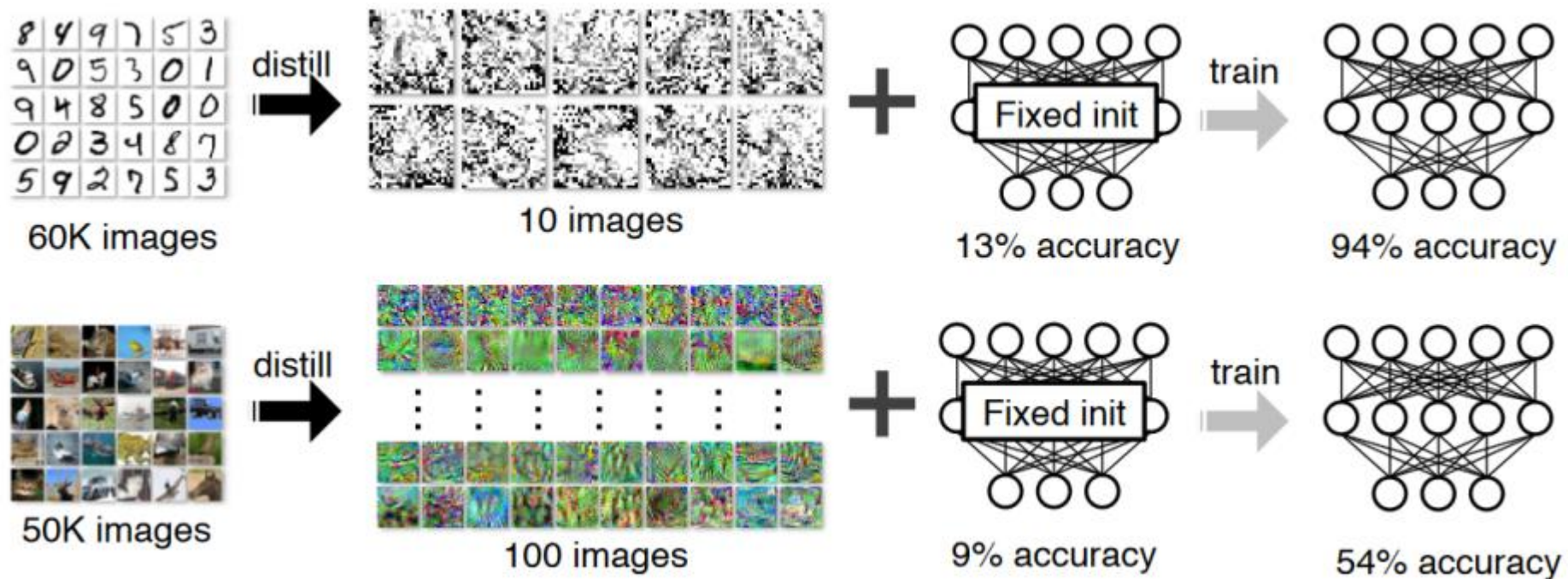
Categories of dataset distillation.

The goal is to ensure the model trained on distilled data performs similarly to the model trained on the full dataset.

The distilled dataset is generated to match the distribution of the original dataset.

The synthetic dataset is optimized to produce similar model parameters as training on the full dataset.

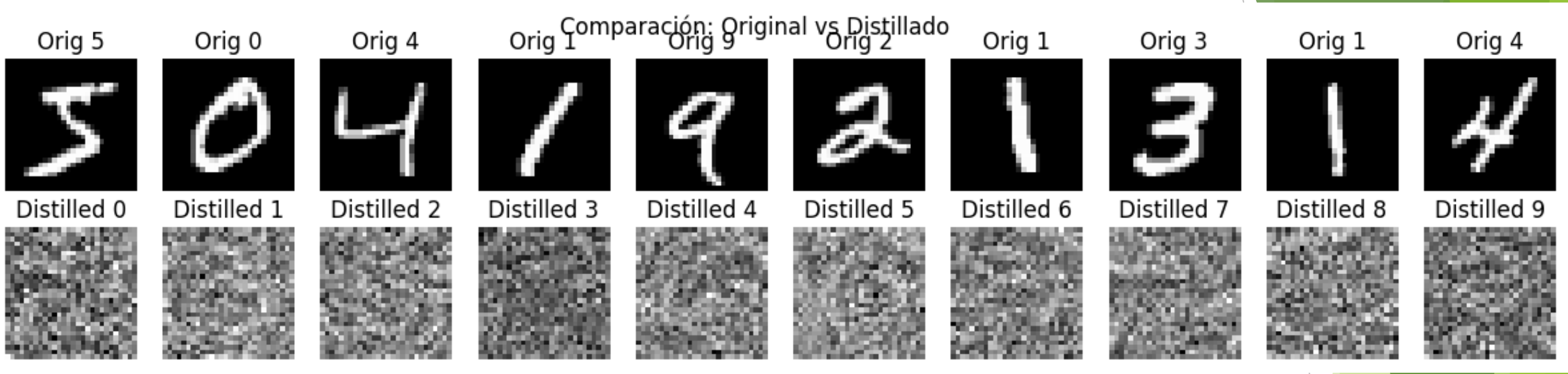
Applications





Code

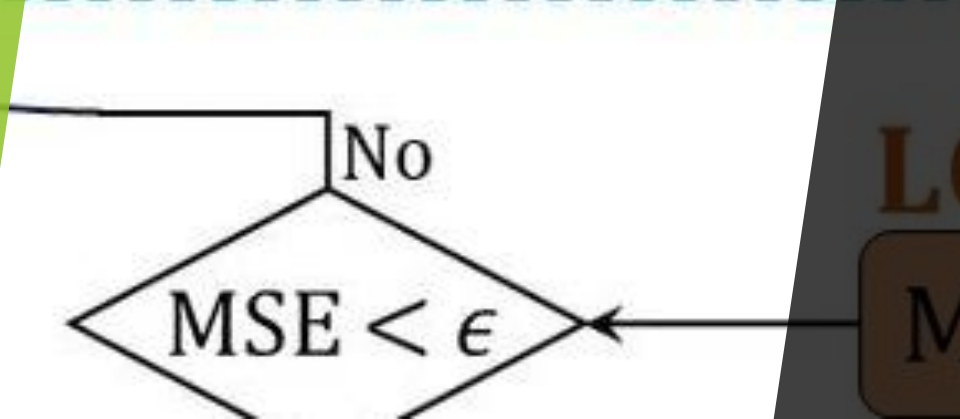
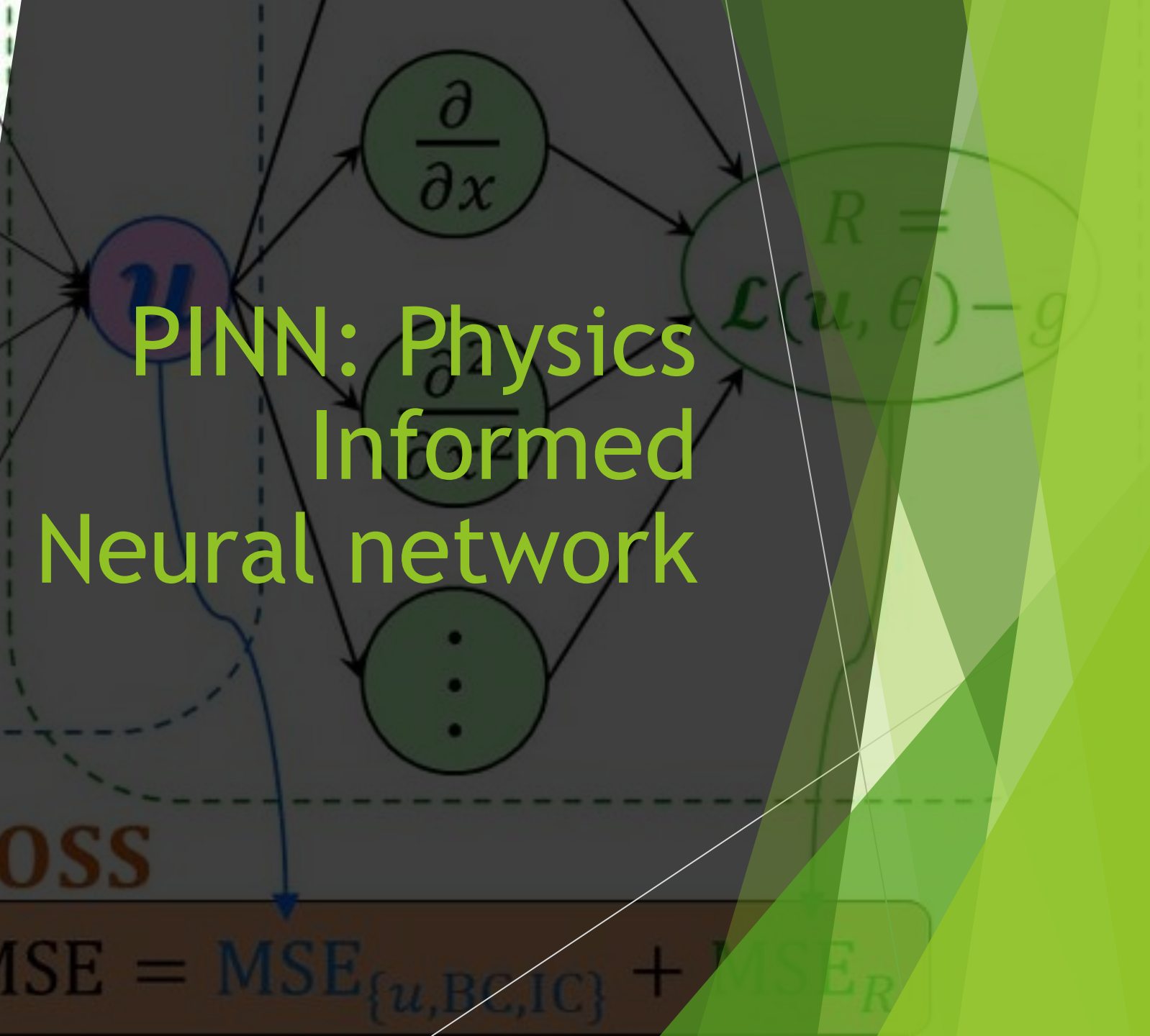
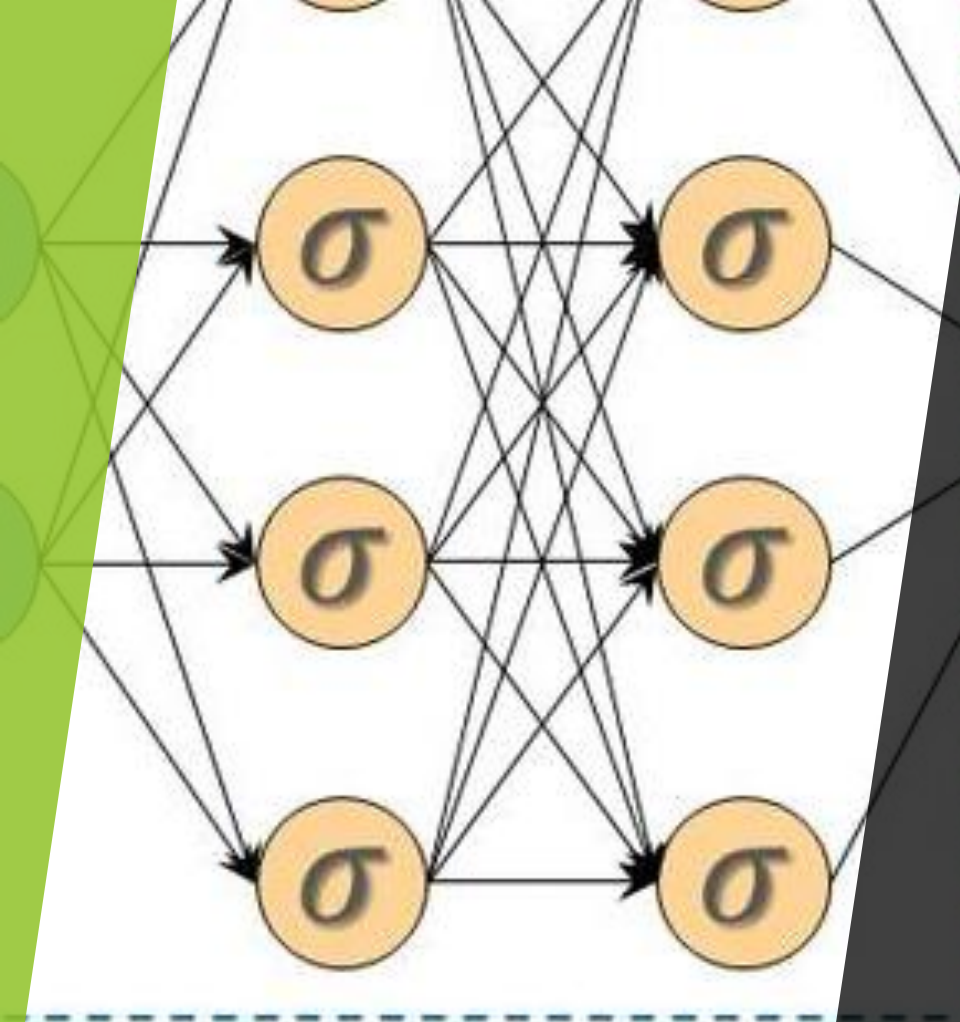
Best Results



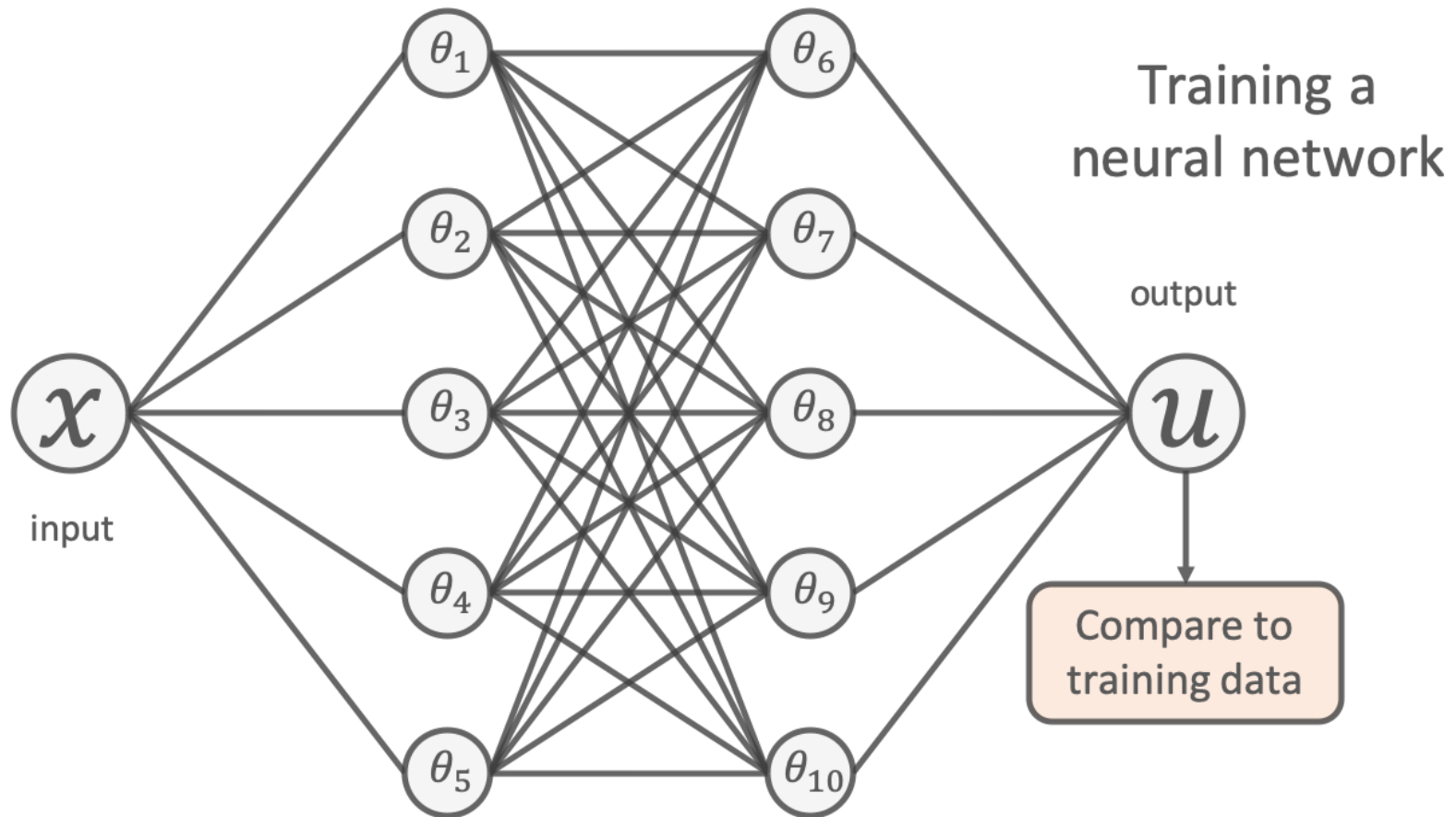
Performance	Accuracy
Teacher	0,957
Student	0,754

Hyperparameters:
Distill images: 200
Epochs Teacher: 500
Epochs Student: 10000
T = 5
Lr_student: 0,001
Dist Steps: 2000

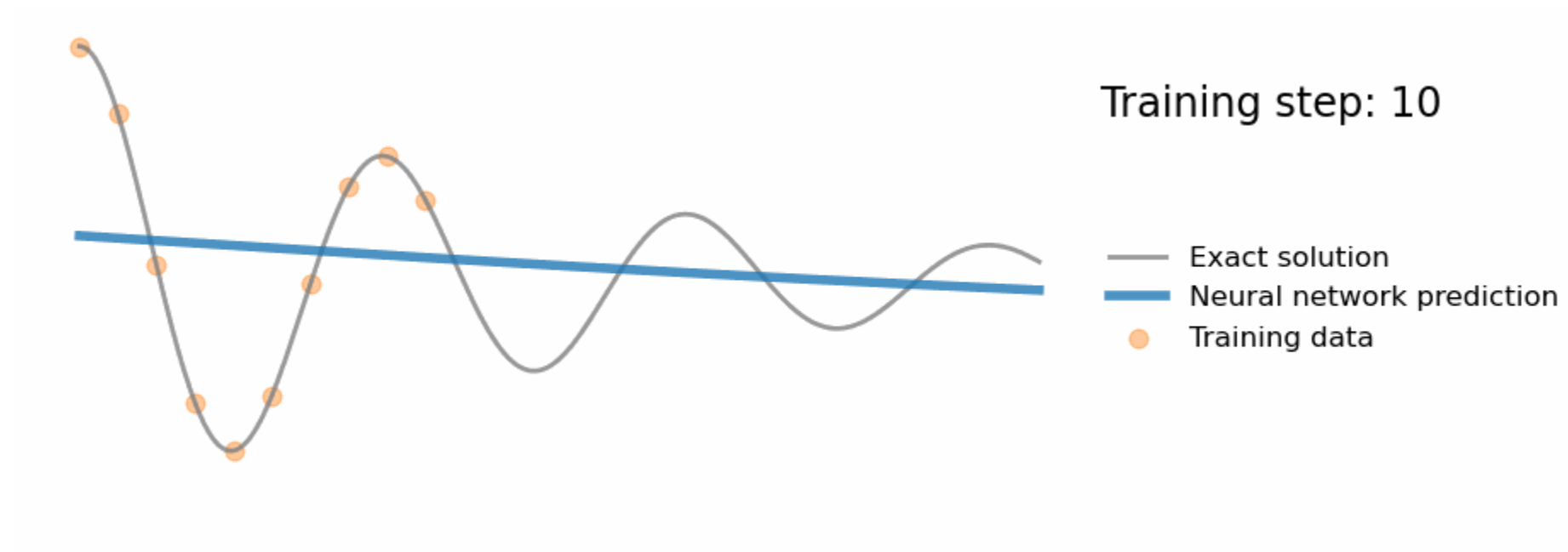
PINN: Physics Informed Neural network



Problem Statement

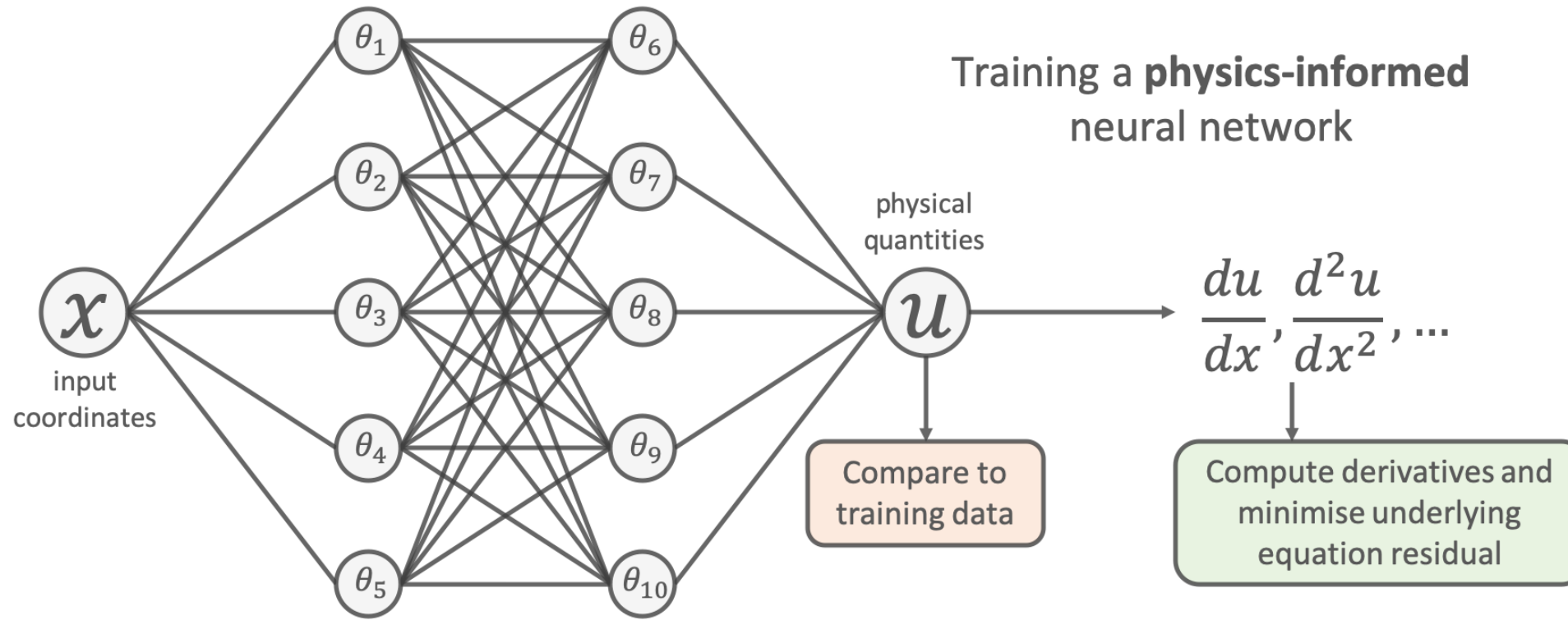


Convergence



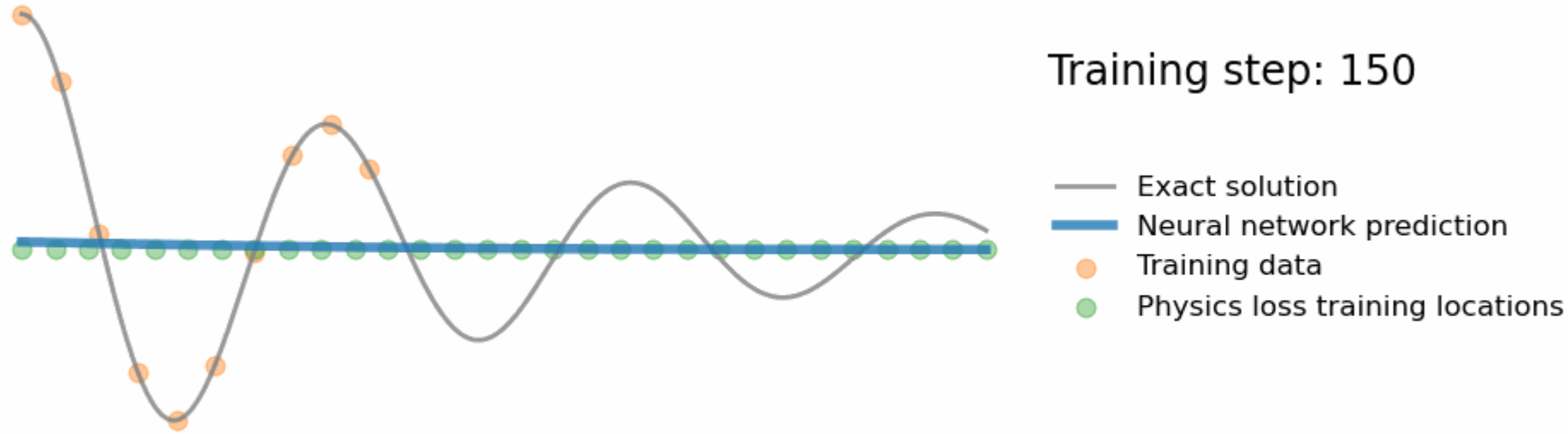
$$Loss = ||u_{NN}(x_i) - u_{true}(x_i)||_2^2$$

Main Concept



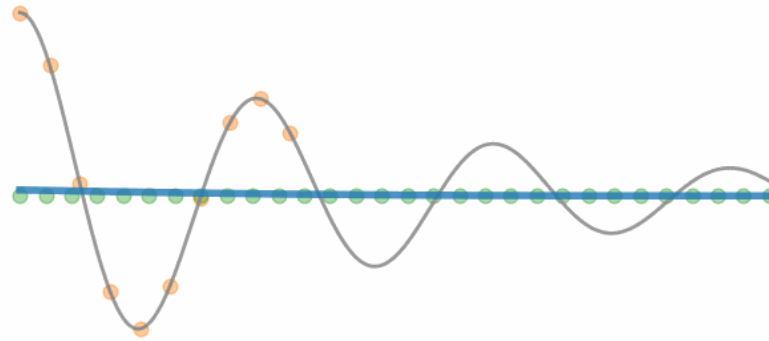
Convergency

Training step: 150



$$Loss = ||u_{NN}(x_i) - u_{true}(x_i)||_2^2 + ||\sum_j \alpha_j \frac{d^j}{dx_i} \{u_{NN}(x_i)\} - f(x_i)||_2^2$$

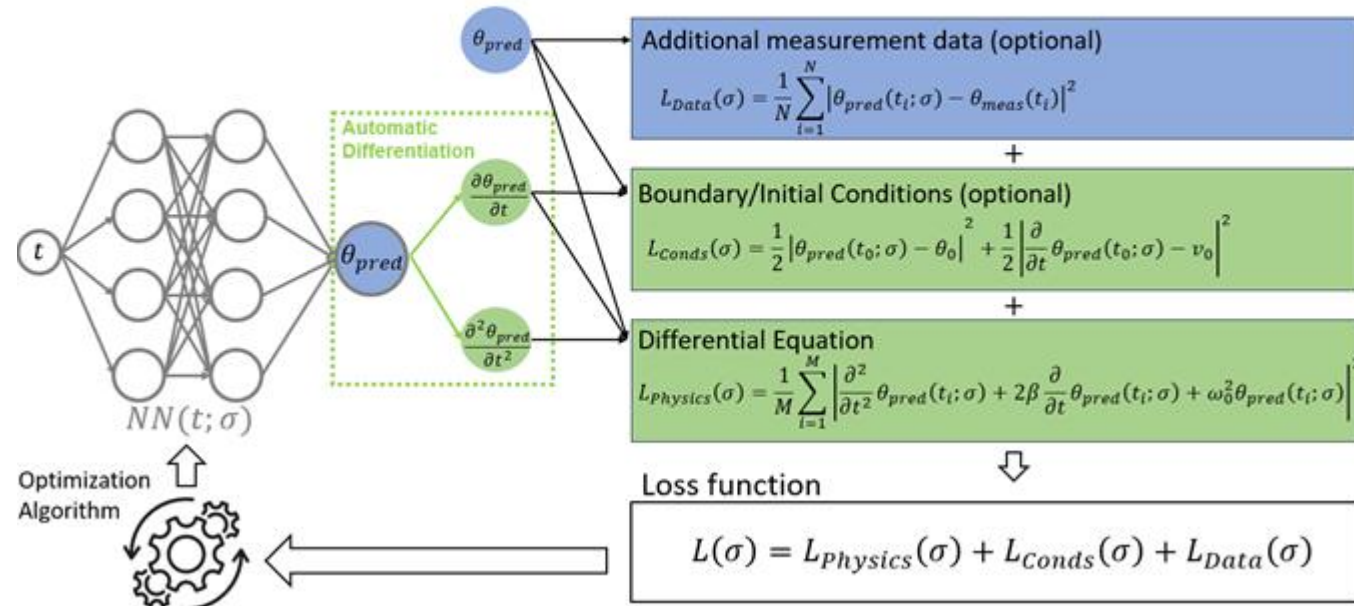
PINN's Loss



Training step: 150

- Exact solution
- Neural network prediction
- Training data
- Physics loss training locations

$$\left. \begin{aligned} \theta''(t) + 2\beta\theta'(t) + \omega_0^2\theta(t) &= 0 \\ \theta(t_0) &= \theta_0 \\ \theta'(t_0) &= v_0 \end{aligned} \right\} \begin{array}{l} \text{Pendulum equation} \\ \text{Initial conditions} \end{array}$$



Applications

PINNs IN ACTION: VALUE FOR THE INDUSTRY

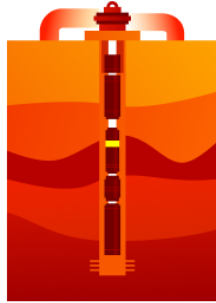


CHEMICAL REACTOR

PINNs run as a part of a hybrid model for a chemical reactor running the Fischer-Tropsch process.

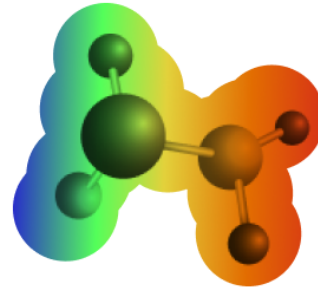
Two PINNs are used to:

- Solve an equation for reaction rates.
- Model the distribution of reactants inside a catalytic pellet.



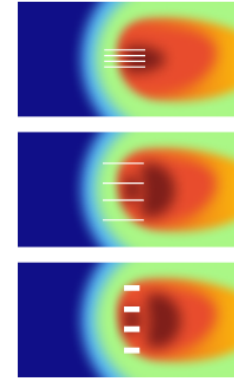
ELECTRICAL SUBMERSIBLE PUMPS

A digital twin of ESP-lifted oil well, based on PINN formalism, takes advantage of combining available sensor data with fluid flow equations, and the approximative power of neural networks.



MOLECULAR ELECTROSTATIC POTENTIAL

Our in-house PINN-based predictive pipeline is designed to produce a spatial distribution of molecular electrostatic potential for a wide range of organic molecules.



HEAT SINK DESIGN

Our PINN-driven solution for heat sink design solves heat transfer equations and optimizes heat sink configuration (e.g., temperature minimization on a CPU).



Code